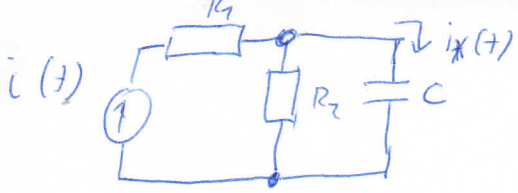

$R_1 = 4\ k\Omega$

$R_2 = 4\ k\Omega$

$C = 1{,}591\ uF$

$I_m = 20\ mA$

$f = 50\ Hz$

$\varphi = \frac{\pi}{3}$

$\omega = 2\pi f = 2\pi\, 50 = 100\pi\ [s^{-1}]$

$i(t) = I_m \sin(\omega t + \varphi) = 0{,}02 \sin\left(100\pi t + \frac{\pi}{3}\right)\ [A]$

$\hat{I} = 0{,}02\, e^{j\frac{\pi}{3}}$

$\hat{I}_x = \hat{I}\, \frac{R_2}{R_2 + \frac{1}{j\omega C}} = 0{,}02\, e^{j\frac{\pi}{3}}\, \frac{4k\Omega}{4k\Omega + \frac{1}{j \cdot 100\pi \cdot 1{,}591 uF}} = \cancel{0{,}017\, e^{j\,0{,}50}}\ [A]$

$0{,}017\, e^{j\,1{,}51}$

$i_x(t) = 0{,}017 \sin(100\pi t + 1{,}51)\ [A]$